

Calling for a bronch at night

The 8 reasons the on-call bronchoscopist gets woken up, what each one looks like, and what to do before you pick up the phone.

Why these 8? In a year-long audit of a 24-hour emergency bronchoscopy service, **614** urgent bronchoscopies were done. **88%** were in the ICU and **84.5%** were therapeutic: collapsed lung, retained secretions, aspiration and airway bleeding. The rest were mostly sampling in immunocompromised patients (Fuehner et al., 2009). Airway emergencies, foreign bodies and inhalation burns complete the list.

01

Lung or lobe collapse from a mucus plug

CALL TONIGHT · the commonest therapeutic night call

You'll see: Failure to wean, rising peak pressures, new desaturation, absent breath sounds on one side. X-ray: lobe or whole lung white, with the trachea and mediastinum pulled **toward** the white side.

Before you call: Check the tube first: a tube in the right main bronchus collapses the **left** lung. Suction, physiotherapy, humidification and turning the patient. Many collapses clear with these alone. Call when they have not worked, or when the patient is getting worse.

Physiotherapy and positioning resolved most atelectasis in one randomised ICU study (Raoof et al., 1999). The bronch is for the plug that will not move.

02

Retained secretions or a partly blocked tube

CALL TONIGHT · ventilated or tracheostomy patients

You'll see: Peak pressures climbing, a suction catheter that will not pass easily, thick or bloody secretions, falling tidal volumes on pressure control.

Before you call: If the tube is **completely** blocked and the patient is crashing, change or remove the tube now. Don't wait for a bronch. For partial blockage: humidify, suction, check tube length and cuff.

03

Aspiration of food or particulate matter

CALL TONIGHT · witnessed vomit, choking on a meal

You'll see: Sudden desaturation after vomiting or eating, new collapse (often right lower lobe), food or debris seen on suctioning.

Before you call: Suction, position and secure the airway. Bronchoscopy is for **solid** material blocking an airway. Liquid gastric aspiration alone does not usually need one.

Antibiotics are not automatic after aspiration. Follow local guidance.

04

Haemoptysis or bleeding in the airway

CALL NOW · massive haemoptysis is a call-now emergency

You'll see: Blood coming up the tube or coughed up, clots blocking the airway, rising pressures, hypoxia.

Before you call: Lie the patient **bleeding side down**. Secure the airway with a large tube (8.0 or bigger, so a scope and suction can pass). Correct clotting, stop anticoagulants, cross-match. Call the bronchoscopist, interventional radiology and thoracic surgery **together**.

In massive haemoptysis the patient dies of asphyxia, not blood loss: control the airway first (Davidson and Shojaee, 2019).

05

Difficult or failed intubation

CALL NOW · anaesthetist-led, scope-assisted

You'll see: Known difficult airway, head and neck tumour or swelling, limited mouth opening or neck movement, a failed laryngoscopy.

Before you call: Oxygenate first, and follow your difficult airway algorithm. A fiberoptic scope can guide intubation (awake or asleep), a tube exchange, or check double-lumen tube position.

Never let the plan for a bronch delay basic oxygenation. Call for help early.

06

Tracheostomy problems

CALL NOW · displaced, blocked or new tracheostomies

You'll see: Loss of capnography trace, a suction catheter that will not pass, surgical emphysema, bleeding from the stoma.

Before you call: Follow your emergency tracheostomy algorithm first (oxygen to face and stoma, remove the inner tube, deflate the cuff). A scope then confirms position or guides a new tube.

07

Foreign body in the airway

CALL TONIGHT · children with nuts or seeds; adults with teeth, dentures or tablets

You'll see: Choking episode, one-sided wheeze or reduced breath sounds, stridor, X-ray showing air trapping or collapse. In adults: elderly, neurological disease, sedation, recent dental work.

Before you call: Keep the patient calm and upright if they are maintaining their airway. Nil by mouth. In a child, call the paediatric airway team early: rigid bronchoscopy may be needed.

Nut or seed exposure, one-sided wheeze, stridor and a suggestive X-ray predicted a foreign body in children (Raviv et al., 2023).

08

Inhalation injury in burns

CALL TONIGHT · house fires, enclosed-space burns

You'll see: Soot in the mouth or sputum, singed nasal hair, facial burns, hoarse voice, stridor, carbon monoxide exposure.

Before you call: Airway first: a hoarse voice and stridor mean the airway can close fast, so intubate early. Bronchoscopy confirms and grades the injury and clears soot and casts.

Can it wait until morning?

Call now	Call tonight	Usually waits for daytime
• Massive haemoptysis • Blocked or displaced tube or tracheostomy • Failed or difficult intubation • Foreign body with breathing difficulty	• Collapse that failed physio and suctioning • Patient getting worse on the ventilator • Solid material aspirated • Inhalation injury needing grading	• Stable collapse still improving with physio • Routine sampling (BAL) in a stable patient • Diagnostic bronchoscopy for a mass • Elective percutaneous tracheostomy

Bronch won't fix it: rule these out first

- **Pneumothorax.** The trachea is pushed **away** from the bad side. Needs a decompression or drain, not a scope. Bedside ultrasound helps.
- **Large pleural effusion.** White lung with the trachea pushed **away**. Needs ultrasound and drainage.
- **Tube in the wrong place.** Check the tube length at the teeth and on the X-ray before calling.
- **Pulmonary oedema, ARDS or pneumonia.** Both sides white or patchy, with no single blocked airway to clear.

What to have ready when you call

- **The story in one line:** why now, and what you've already tried.
- **Latest chest X-ray** (and CT if done). Ventilator settings, FiO₂, PEEP, peak pressures, last blood gas.
- **Tube size and length.** An adult scope needs a tube of 8.0 or bigger to ventilate around it. Below 7.5, warn the bronchoscopist.
- **Platelets, clotting and anticoagulants**, especially if a biopsy or bleeding is possible.
- **Sedation and paralysis plan.** The patient will be pre-oxygenated on FiO₂ 1.0, so have a nurse and drugs ready.
- **Consent** or best-interests documentation, and the family informed.
- **Samples:** know what you want sent (cultures, cytology, fungal or TB studies) before the scope goes in.

References

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